MealSync

Smart Campus Food Management

User Guide

Predict Smart. Reduce Waste. Share More.

What is MealSync?

MealSync helps a campus mess hall cook the right amount of food, track every cooked batch, and safely pass surplus food to approved collection points such as the student welfare centre or food bank. It runs entirely in your web browser; no account or server is needed.

This guide covers

Section	What you will learn
1. Getting started	Opening the app and finding your way around
2. Pages at a glance	What each of the 7 pages is for
3. Everyday tasks	Add, verify, edit and redistribute a batch
4. Food safety rules	The rules the app enforces and why
5. Forecast, analytics, settings	Reading and using the planning tools
6. Your data and privacy	Where information is stored
7. Troubleshooting & FAQ	Fixes for common problems

> **Before you begin**
> Use a current browser (Chrome, Edge, Firefox or Safari) and an internet connection the first time you open the page, because styling and charts load from the web. Do not use private/incognito mode if you want your data to be remembered.

1. Getting started

1. Open the MealSync file (*.html*) in your browser, or visit the address your administrator gave you.
2. The **Dashboard** opens first. You are signed in as the demo operator, **Elena Vance, Campus Dining Lead**.
3. Use the **top navigation bar** to move between pages. On a phone, tap the menu button to show the same links.
4. Press **Esc** at any time to close a pop-up window or dropdown menu.

The top bar

Item	What it does
Skip to main content	Press **Tab** once on page load; it lets keyboard users jump past the menu.
Green “Local IndexedDB Storage” badge	Opens the privacy notice explaining that data stays in your browser.
Bell icon	Shows operator safety and batch alerts (sample alerts in this version).
Profile pill (Elena Vance)	Opens the operator menu.

Colour and labels

Every status carries a **text label** as well as a colour (for example “[Safety Verified]” or “[Unverified - Not Eligible]”), so the app can be read without relying on colour alone.

2. Pages at a glance

Page	Purpose
Dashboard	Today’s demand, meals consumed, potential surplus and meals redistributed; a table of the latest batches with quick action buttons; the **Add Food Batch** button.
Inventory	Full list of every batch with search and filters (counter, status, dietary category). Verify, edit or redistribute from here.
Surplus Detection	Card view of surplus batches with quick filters: All, Pending Verification, Verified Safe, plus counter, dietary and minimum-quantity filters.
Demand Forecast	Attendance-based kitchen prep targets, a forecast chart and a what-if simulator.
Redistribution	Pick a verified batch, see which collection points can accept it, confirm the handoff and review the audit log.
Analytics	Charts and totals for food prepared, consumed, saved; export an audit report.
Settings	Campus information, counters, collection points, food handling policy and notification preferences.

3. Everyday tasks

3.1 Add a food batch

1. Click **Add Food Batch** (Dashboard or Inventory). A form opens with a new unique **Batch ID**.
2. Enter the **Food item name**, choose the **Cafeteria counter**, and enter the **Quantity** (1 to 500 meals).
3. Check the **prepared time** and the **approved handling deadline** (they default to now and four hours from now).
4. Choose the **Dietary classification** (Vegetarian, Vegan or Non-Vegetarian) and tick every **allergen** that applies: Gluten, Dairy, Peanuts, Tree Nuts, Soy. Tick *None* only if there are no allergens.
5. Click save. The batch appears with status **Pending Verification**.

> **Tip**
> The “verified safe” tick box in the add form is for authorised operators who have already inspected the food. If you are unsure, leave it unticked and verify later.

3.2 Verify a batch’s safety

A batch cannot be redistributed until an operator has physically checked it. Click **Verify Batch Safety** on the batch (Dashboard, Inventory or Surplus page). Confirm all four checkpoints:

Checkpoint	Meaning
Temperature	Food has been held at a safe temperature
Sensory check	Appearance, smell and texture are normal
Packaging	Containers are clean, sealed and labelled
Handling window	The approved deadline has not passed

All four must be ticked, otherwise the app shows an error and the batch stays unverified. After approval the status changes to **Available** and your name is recorded as the verifier.

3.3 Edit a batch

Click **Edit** on a batch in Inventory (or on the Dashboard for batches without another action). You can change the food name, counter, quantity, handling deadline, dietary category and verified status. **Un-ticking “verified”** withdraws the safety sign-off and sends the batch back to *Pending Verification*.

3.4 Redistribute surplus food

1. Go to **Redistribution** (or click **Initiate Redistribution** on a verified batch). The left list shows candidate batches.
2. Click a **verified** batch. Unverified batches show “Safety Blocked” and offer a **Verify Batch Safety Now** button.
3. The right side lists every collection point. Points marked **[Recommended Match]** can accept the batch; **[Incompatible]** ones show the reason (not enough capacity, dietary mismatch or allergen restriction).
4. Click **Confirm Redistribution** on a recommended point and review the summary, then confirm.
5. The batch becomes **Collected**, the point’s reserved capacity goes up, “meals saved” increases, and a new line is added to the **Handoff Audit Log**.

4. Food safety rules

> **Core rule**
> MealSync never decides on its own that food is safe. A batch is eligible only after an authorised operator verifies it and confirms the approved handling window is valid.

- **Unverified batches are blocked** from matching and from confirmation.
- **Expired Window** batches cannot be redistributed; they are marked for discard or composting.
- **Capacity is checked** at confirmation, so a collection point cannot be over-booked.
- **Dietary rules:** the Approved Community Kitchen accepts Vegetarian and Vegan only.
- **Allergen rules:** collection points list allergens they cannot accept (for example Peanuts, Tree Nuts, Shellfish) and are excluded automatically.
- Every transfer is recorded with batch, quantity, destination, verifier and time.

Default collection points

Point	Capacity	Notes
Student Welfare Centre	35 meals	Does not accept Peanuts or Tree Nuts
Campus Food Bank	50 meals	Accepts all
Approved Community Kitchen	25 meals	Vegetarian & Vegan only; no Peanuts, Tree Nuts, Shellfish
Hostel Collection Point	20 meals	Veg and Non-Veg; open 24 hours

Batch statuses

Status	Meaning
Pending Verification	Waiting for an operator safety check
Available	Verified and eligible for redistribution
Matched	Reserved for a collection point
Collected	Handed over and logged
Expired Window	Handling deadline missed; discard protocol

5. Forecast, analytics and settings

Demand Forecast

- Choose **Today**, **Last 7 Days** or **Last 30 Days** to switch the summary figures.
- **Simulate Today's Morning Turnstile Swipes** updates the "actual" count with sample check-ins.
- **Apply Target to Today's Prep Plan** confirms the recommended cook target (formula: 0.5 × yesterday + 0.3 × day-2 + 0.2 × day-3, plus a 5% buffer).
- Change the day, event or weather options and click **Recalculate Dynamic Estimate** to see a what-if demand and variance.

Analytics

Pick a date range, read the four charts (predicted vs actual, waste trend, share by counter, cumulative meals saved) and click **Export Audit Report** to download a **CSV** of all redistribution transactions, which opens in Excel or Google Sheets.

Settings

Settings starts in **Operator View (Read-Only Policy)**. Click **Switch to Admin Role** to unlock policy changes such as the maximum handling window, hot/cold holding temperatures, and adding counters or collection points. Without admin mode, saving shows “Permission Denied”.

6. Your data and privacy

- All batches, collection points and transfer history are stored **locally in your browser** (IndexedDB). Nothing is sent to a server.
- Data stays on **this device and browser only**. Another computer or browser will start with fresh demo data.
- Clearing your browser’s site data erases MealSync data.
- If the browser cannot provide local storage (for example private mode), the app runs in **temporary mode** and shows a yellow notice: changes are lost on reload.
- To start over, open the **profile menu** (Elena Vance, top right) and choose the reset-demo-data option; it restores the sample batches and history.

> **Demo environment**
> Sample figures, timestamps, alerts, forecast and analytics numbers are demonstration data for the campus pilot. Only batches, safety sign-offs, redistribution transfers and meals-saved totals change as you use the app.

7. Troubleshooting & FAQ

Problem	What to do
Page looks unstyled or charts are empty	Check your internet connection and reload; styling and charts load from the web.
I can’t click Confirm Redistribution	The batch must be verified and the point must have enough capacity. Verify the batch or choose another point.
No collection point is recommended	The batch is too large or its dietary/allergen tags conflict with every point. Reduce quantity (Edit) or wait for capacity.
Edit form won’t save	Make sure the name is filled, quantity is 1–500 and a deadline time is set.
My data disappeared	You may have cleared browser data, used private mode, or opened another browser. Data is stored per browser.
“Permission Denied” in Settings	Click Switch to Admin Role first.
A batch is stuck as Expired Window	Expired batches cannot be redistributed by design; follow your campus discard protocol.

Keyboard shortcuts

Key	Action
Tab / Shift+Tab	Move between buttons and fields
Enter or Space	Activate a button or select a batch card
Esc	Close pop-ups and dropdowns

Need help? Contact your campus dining administrator or the Campus Safety Hotline listed under Settings → Campus Information.